

Battery Specifications

Executive Summary

Nominal Voltage	51.2V
Total Capacity	42Ah (≈ 2150Wh)
System Max Continuous Discharge	60A(LimitedbyBMS)
Cell-Level Max Continuous Discharge	126A
Cell Chemistry	Lithium Iron Phosphate (LiFePO ₄)
Expected Life Cycle	> 2000Cycle
Series/Parallel Configuration	16S7P
Cooling Method	Passive

System Requirements

The battery pack is designed to power a BLDCM with the following key specifications:

Nominal Voltage	48V
Rated Speed	3000 RPM
Rated Power	660W

The system requires two hours of continuous operation at full load. Assuming a combined electromechanical conversion efficiency (inverter and motor) of 80%, the required energy capacity is calculated as follows:

$$E_{\text{req}} = \frac{660W}{80\%} \times 2h \approx 1650 \text{ Wh}$$

Consequently, the required capacity in Ampere-hours (Ah) at the nominal system voltage is:

$$Ah_{\text{req}} = \frac{1650 \text{ Wh}}{48V} \approx 34.38 \text{ Ah}$$

Battery Design

Chemistry Selection

Lithium Iron Phosphate (LiFePO₄) was selected based on the following engineering parameters:

1. **Thermal Stability:** Inherently stable chemistry minimizes the risk of thermal runaway events.
2. **Cycle Life:** Capable of > 2000 *charge/discharge cycles*, providing a longer operational lifespan compared to alternative lithium chemistries.
3. **Environmental Impact:** Free of toxic heavy metals.
4. **Environmental Tolerance:** Maintains reliable performance across a wide operating temperature range.
5. **Total Cost of Ownership:** Extended lifespan and reduced maintenance requirements yield long-term cost efficiency.

Cell Specifications

Form Factor	32700
Nominal Voltage	3.2V
Nominal Capacity	6 Ah
Max Continuous Discharge Current	18 A

Electrical Design

To size the battery pack appropriately, the required capacity must account for the safe Depth of Discharge (DoD) limit of the cells. Assuming a strict DoD limit of 90% (0.9) to preserve cell longevity:

$$Ah_{pack} = \frac{Ah_{req}}{DoD_{limit}}$$
$$Ah_{pack} = \frac{34.38 \text{ Ah}}{0.9} = 38.2 \text{ Ah}$$

Parallel Configuration (N_p):

Given each cell's capacity of 6 Ah, the required number of parallel cells is:

$$N_p = \frac{38.2}{6} = 6.37$$

Rounding up to ensure adequate capacity, N_p is designated as 7.

Series Configuration (N_s):

The motor's nominal voltage requirement is 48 V. Based on the 3.2 V nominal voltage of each cell:

$$N_{s\ (base)} = \frac{48}{3.2} = 15$$

While 15S yields exactly 48V, a 16S configuration is standard for 48V-class systems. Increasing N_s to 16 accounts for voltage sag under full load, compensates for resistive losses across system wiring and inverter switching.

Final Pack Specifications (16S7P):

Nominal Voltage	$16 \times 3.2\text{ V} = 51.2\text{ V}$
Nominal Capacity	$7 \times 6\text{ Ah} = 42\text{ Ah}$ (yielding $\approx 2150\text{ Wh}$)
Max Continuous Discharge Current	$7 \times 18\text{ A} = 126\text{ A}$

Implementation Details

The final physical assembly utilizes the designed 16S7P configuration using EPRO 32700 LiFePO4 cells.



Battery Management System (BMS):

The pack is governed by a JIABAIDA DB24SA03 BMS.

- **Hardware Capabilities:** Supports up to 24 series strings.
- **Current Limiting:** Features a maximum continuous current rating of 60 A. **Note:** This acts as the primary system bottleneck; the pack cannot output the cell-capable 126 A without triggering over-current protection. However, 60 A is highly sufficient for the continuous ≈ 18 A draw of the target load.
- **Telemetry & Communication:** Features Bluetooth connectivity for mobile application monitoring and a UART port for external communication protocols.

